

数学练习 第 01 份 答案与解析

共 75 题 与《练习01》题号一一对应

1. A: 取 $M = \frac{1+a}{2} > 1$ 即可, 正确. B: 取 $\varepsilon = \frac{b-a}{2}$, 正确. C: 极限保序性, 正确. D: 反例 $a_n = a - \frac{1}{n}$, 此时 $a_n > a - \frac{1}{n}$ 不成立. 故错误的是 D.

2. 由通解知 $r(A) = 3 - 1 = 2$, 且

$$\alpha_1 + 2\alpha_2 - \alpha_3 = \beta, \quad \alpha_1 - 2\alpha_2 + 3\alpha_3 = 0.$$

于是 $\beta + \alpha_3 = \alpha_1 + 2\alpha_2$, 故 $r(B) = r(\alpha_1, \alpha_2, \alpha_3) = 2$, $By = 0$ 的基础解系含 $4 - 2 = 2$ 个向量.

设 $y = (y_1, y_2, y_3, y_4)^T$, 则

$$(y_1 + y_4)\alpha_1 + (y_2 + 2y_4)\alpha_2 + y_3\alpha_3 = 0 \implies (y_1 + y_4, y_2 + 2y_4, y_3) = t(1, -2, 3),$$

取 $t = 1, y_4 = 0$ 与 $t = 0, y_4 = 1$ 得基础解系 $(1, -2, 3, 0)^T, (-1, -2, 0, 1)^T$.

又 $\alpha_1 - \alpha_2$ 显然由 $y = (1, -1, 0, 0)^T$ 给出特解, 故

$$y = (1, -1, 0, 0)^T + k_1(1, -2, 3, 0)^T + k_2(-1, -2, 0, 1)^T.$$

3. 记 $P = \frac{-y}{x^2+y^2}, Q = \frac{x}{x^2+y^2}$, 除原点外 $\frac{\partial Q}{\partial x} = \frac{\partial P}{\partial y}$.

(I) 圆 $(x+2)^2 + (y-2)^2 = 1$ 不含原点, 由格林公式 $I = 0$.

(III) 椭圆内含原点, 取小圆 $x^2 + y^2 = \varepsilon^2$ 挖去, 得

$$I = \oint_{x^2+y^2=\varepsilon^2} \frac{x dy - y dx}{\varepsilon^2} = \frac{1}{\varepsilon^2} \cdot 2 \cdot \pi \varepsilon^2 = 2\pi.$$

4. 由 $(E - A)(E + A + \cdots + A^{n-1}) = E - A^n = E$ 得

$$(E - A)^{-1} = E + A + A^2 + \cdots + A^{n-1}.$$

5. (I) $r^2 = 2 \cos 2\theta$ 的区域关于 x 轴对称, 而 xy 关于 y 为奇函数, 故 $I = 0$.

(II) $r^2 = 2 \sin 2\theta$, 两叶分别在一、三象限, 关于原点对称, xy 在此变换下不变, 故

$$I = 2 \int_0^{\frac{\pi}{2}} \int_0^{\sqrt{2 \sin 2\theta}} r^3 \sin \theta \cos \theta dr d\theta = 2 \int_0^{\frac{\pi}{2}} \frac{\sin 2\theta}{2} \cdot \frac{(2 \sin 2\theta)^2}{4} d\theta = \int_0^{\frac{\pi}{2}} \sin^3 2\theta d\theta = \frac{2}{3}.$$

6. 记 $A = \int_0^{\frac{1}{2}} f(x) dx$, 则 $f(x) = \frac{1}{(1+x)^2} - 2A$, 两边在 $[0, \frac{1}{2}]$ 上积分:

$$A = \left[-\frac{1}{1+x} \right]_0^{\frac{1}{2}} - A = \frac{1}{3} - A \implies A = \frac{1}{6}, \quad f(x) = \frac{1}{(1+x)^2} - \frac{1}{3}.$$

由 $\frac{1}{(1+x)^2} = \sum_{n=0}^{\infty} (-1)^n (n+1) x^n$ ($|x| < 1$) 得

$$f(x) = \frac{2}{3} + \sum_{n=1}^{\infty} (-1)^n (n+1) x^n, \quad -1 < x < 1.$$

7. X 服从柯西分布, $F_X(x) = \frac{1}{2} + \frac{1}{\pi} \arctan x$. 由 $Y = 1 - \sqrt[3]{X}$ 单调减,

$$F_Y(y) = P\{X \geq (1-y)^3\} = 1 - F_X((1-y)^3) = \frac{1}{2} - \frac{1}{\pi} \arctan(1-y)^3,$$

$$f_Y(y) = \frac{3(1-y)^2}{\pi[1+(1-y)^6]}, \quad -\infty < y < +\infty.$$

8. 令 $u = \frac{y}{x}$, 则 $g_x = -\frac{y}{x^2}f'$, $g_{xx} = \frac{y^2}{x^4}f'' + \frac{2y}{x^3}f'$, $g_{xy} = -\frac{y}{x^3}f'' - \frac{1}{x^2}f'$, $g_{yy} = \frac{1}{x^2}f''$. 代入得

$$x^3 g_{xx} + x^2 y g_{xy} + x y^2 g_{yy} = \frac{y^2}{x} f'' + y f' = y \implies u f''(u) + f'(u) = 1.$$

即 $(u f')' = 1$, $u f' = u + C$, $f'(u) = 1 + \frac{C}{u}$, $f(u) = u + C \ln u + D$.

由 $g(y, y) = f(1) = 1$ 得 $D = 0$; 由 $\frac{\partial g}{\partial y}|_{(y, y)} = \frac{f'(1)}{y} = \frac{2}{y}$ 得 $f'(1) = 2$, $C = 1$. 故

$$f(t) = t + \ln t.$$

9. $EX = DX = 1$.

$$DY_i = D \left[\left(1 - \frac{1}{n}\right) X_i - \frac{1}{n} \sum_{j \neq i} X_j \right] = \left(1 - \frac{1}{n}\right)^2 + \frac{n-1}{n^2} = 1 - \frac{1}{n},$$

$T = (n-1)S^2$, $ET = (n-1)DX = n-1$. 选 B.

10. ①错: 取 $f(a+t) - f(a) = |t|^{2/3}$, 则 $\frac{f(a+x^3) - f(a)}{x^2} \equiv 1$ 极限存在, 但 $f'(a)$ 不存在.
 ②错: 数列极限存在不能推出函数极限存在, 取 f 在 $x = \frac{1}{n}$ 及 0 处取 0、其余取 1, 则该极限为 0, 但 f 在 $a = 0$ 处不连续.
 ③对: $f'(a) = \lim_{x \rightarrow a} \frac{(x-a)\varphi(x)}{x-a} = \varphi(a)$.
 故仅 ③ 正确 (原书选项缺失, 按“仅③正确”作答).

11. 设 $s = x + y$, $p = xy$, 由 $x^2 + y^2 + xy = 3$ 得 $p = s^2 - 3$, $x^2 + y^2 = 6 - s^2$, 于是

$$f = (1+x)^2 + (1+y)^2 = x^2 + y^2 + 2s + 2 = 8 + 2s - s^2 = 9 - (s-1)^2.$$

x, y 为实数要求 $s^2 \geq 4p$, 即 $s^2 \leq 4$, $s \in [-2, 2]$. 取 $s = 1$ 得最大值

$$f_{\max} = 9 \quad (\text{此时 } x = 2, y = -1 \text{ 或 } x = -1, y = 2).$$

12. (I) 似然函数 $L = \theta^{-n} (0 \leq x_{(n)} \leq \theta)$ 关于 θ 单调减, 故 $\hat{\theta} = X_{(n)} = \max\{X_1, \dots, X_n\}$.
 (II) $X_{(n)}$ 的密度为 $\frac{nx^{n-1}}{\theta^n} (0 \leq x \leq \theta)$, 故

$$E\hat{\theta} = \frac{n}{n+1}\theta, \quad E\hat{\theta}^2 = \frac{n}{n+2}\theta^2, \quad D\hat{\theta} = \frac{n\theta^2}{(n+2)(n+1)^2}.$$

13. 在 L 上 $x^2 + y^2 = 2x$, $z^2 = x^2 + y^2 = 2x$, $y^2 = 2x - x^2$, 且 $z = \sqrt{2x}$, $dz = \frac{dx}{z}$, 于是

$$(y^2 + z^2)dx + (x^2 + y^2)dz = (4x - x^2)dx + \sqrt{2x} dx$$

为只含 x 的全微分, 沿闭曲线积分为 0. 故
 由格林公式

$$I = \oint_L (2x - x^2) dy = \iint_{x^2 + y^2 \leq 2x} (2 - 2x) d\sigma = 2\pi - 2\pi = 0.$$

14. (I) $f_X(x) = \int_x^{+\infty} x e^{-y} dy = x e^{-x} (x > 0)$, $f_Y(y) = \int_0^y x e^{-y} dx = \frac{y^2}{2} e^{-y} (y > 0)$.
 因 $f(x, y) \neq f_X(x) f_Y(y)$, 故 X, Y 不独立.
 (II) 当 $0 < x \leq y$ 时 $F(x, y) = 1 - (1+x)e^{-x} - \frac{x^2}{2} e^{-y}$;
 当 $0 < y < x$ 时 $F(x, y) = 1 - (1+y + \frac{y^2}{2})e^{-y}$; 其余为 0.
 (III) $f_Z(z) = \int_0^{z/2} x e^{-(z-x)} dx = \left(\frac{z}{2} - 1\right) e^{-z/2} + e^{-z} (z > 0)$, 其余为 0.

15. 被积函数 $= \frac{(a-b)x+b}{x(2x+b)} = \frac{1}{x} + \frac{a-b-2}{2x+b}$. 积分收敛要求 $1 + \frac{a-b-2}{2} = 0$, 即 $a = b$. 此时

$$\int_1^{+\infty} \left(\frac{1}{x} - \frac{2}{2x+b} \right) dx = \left[\ln \frac{x}{2x+b} \right]_1^{+\infty} = \ln \frac{2+b}{2} = 1 \implies b = 2(e-1) = a.$$

选 B.

16. (I) $|\lambda E - A| = \lambda(\lambda-1)(\lambda-(2+a^2))$, 特征值 $0, 1, 2+a^2$. $A - kE$ 半负定 $\iff k \geq \lambda_{\max}$, 故 $k_{\min} = 2+a^2$.

(II) 配方:

$$f = x_1^2 + (1+a^2)x_2^2 + x_3^2 - 2x_1x_2 + 2ax_2x_3 = (x_1-x_2)^2 + (ax_2+x_3)^2,$$

而 $g = (y_1-y_2)^2 + y_3^2$. 取 $x_1-x_2 = y_1-y_2$, $x_2 = y_2$, $ax_2+x_3 = y_3$, 即

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -a & 1 \end{pmatrix} \quad (|C| = 1 \neq 0),$$

则 $X = CY$ 将 f 化为 g .

17. 由 $AB = E_m$ 知 $r(A) = r(B) = m$. A 为 $m \times n$ 行满秩, 故 $AX = \alpha$ 必有解 (未必唯一); B 为 $n \times m$ 列满秩, 故 $BX = 0$ 只有零解 (而 $BX = \beta$ 未必有解). 选 D.

18. $\sqrt[3]{1-x^6} = -x^2 \left(1 - \frac{1}{x^6} \right)^{1/3} = -x^2 + \frac{1}{3x^4} + o(x^{-4}),$

故 $a = -1, b = 0$.

19. $\rho = 0$ 故 X, Y 独立, $EX = EY = 1$, $EX^2 = 2+1=3$, $EY^2 = 4+1=5$.

$$D(XY) = E(X^2Y^2) - (EXEY)^2 = 3 \times 5 - 1 = 14.$$

20. (I) 收敛域 $[-2, 2)$, $S(x) = -\frac{\ln(1-\frac{x}{2})}{x}$ ($x \neq 0$), $S(0) = \frac{1}{2}$.

(II) 收敛域 $(-1, 1)$. 由 $\frac{(n-1)^2}{n+1} = n-3 + \frac{4}{n+1}$ 得

$$S(x) = \frac{x}{(1-x)^2} - \frac{3}{1-x} - \frac{4\ln(1-x)}{x} \quad (x \neq 0), \quad S(0) = 1.$$

(III) 收敛域 $(-\infty, +\infty)$. 由 $n^2+1 = n(n-1) + n+1$ 得 $S(x) = (x^2+x+1)e^x$.

(IV) 收敛域 $(-\infty, +\infty)$. 由 $2(n+1) = (2n+3) - 1$ 得

$$x^3 S(x) = \frac{1}{2} [x(1-\cos x) - (x-\sin x)] = \frac{\sin x - x \cos x}{2}, \quad S(x) = \frac{\sin x - x \cos x}{2x^3} \quad (x \neq 0), \quad S(0) = \frac{1}{6}.$$

21. 伯努利方程. 令 $z = y^{-1}$, 得 $z' + \frac{6}{x}z = x$, 积分因子 $x^6: (x^6z)' = x^7, z = \frac{x^2}{8} + \frac{C}{x^6}$. 故

$$\frac{1}{y} = \frac{x^2}{8} + \frac{C}{x^6}, \quad \text{即} \quad y = \frac{8x^6}{x^8 + C_1} \quad (C_1 = 8C).$$

22. $|B| = -|A| = -2$, 而

$$|B^{-1}B^*B^T| = \frac{1}{|B|} \cdot |B|^{n-1} \cdot |B| = |B|^{n-1} = (-2)^{n-1}.$$

23. $r(A) = 1$ 知 $\lambda = 0$ 是二重特征值, $AX = 0$ 的解空间即 $\lambda = 0$ 的特征子空间, 与 $\alpha = (-1, 1, 1)^T$ 正交. 验证 $(1, 1, 0), (1, 0, 1)$ 均与 α 正交且线性无关. 选 B.

24. 设最小值点为 $c \in (0, 1)$, 则 $f'(c) = 0$ 且 $f(c) \leq f(0) = 0$.

(I) 当 $f(c) = f(0) = 0$ 时, 由罗尔定理存在 $\xi_1 \in (0, c)$ 使 $f'(\xi_1) = 0$; 再对 f' 在 $[\xi_1, c]$ 上用罗尔定理得 $\xi \in (\xi_1, c) \subset (0, 1)$ 使 $f''(\xi) = 0$.

(II) 令 $F(x) = e^{-\lambda x} f'(x)$, 则 $F(\xi_1) = F(c) = 0$, 由罗尔定理存在 $\eta \in (\xi_1, c)$ 使

$$F'(\eta) = e^{-\lambda \eta} [f''(\eta) - \lambda f'(\eta)] = 0 \implies f''(\eta) - \lambda f'(\eta) = 0.$$

注: 若最小值 $f(c) < f(0)$, 结论 (I) 不成立 (反例 $f(x) = 2(x - \frac{1}{4})^2 - \frac{1}{8}$ 满足 $f(0) = 0, f(1) = 1$ 而 $f'' \equiv 4$), 故本题应按“最小值 $= f(0) = 0$ ”理解, 题干请核对.

25. 两边分别对 x, y 求偏导:

$$\left(1 + \frac{1}{z}\right)z_x = e^{-x^2}, \quad \left(1 + \frac{1}{z}\right)z_y = -e^{-y^2} \implies z_x = \frac{ze^{-x^2}}{1+z}, \quad z_y = -\frac{ze^{-y^2}}{1+z}.$$

故

$$\frac{\partial^2 z}{\partial x \partial y} = e^{-x^2} \frac{\partial}{\partial y} \left(\frac{z}{1+z} \right) = \frac{e^{-x^2} z_y}{(1+z)^2} = -\frac{ze^{-x^2-y^2}}{(1+z)^3}.$$

26. $F_X(x) = x^3$ ($0 < x < 1$). 当 $y > 1$ 时

$$F_Y(y) = P\left\{X \geq \frac{1}{y}\right\} = 1 - \frac{1}{y^3},$$

$y \leq 1$ 时 $F_Y(y) = 0$. 故 $f_Y(y) = \frac{3}{y^4}$ ($y > 1$), 其余为 0.

27. 极坐标: $0 \leq \theta \leq \frac{\pi}{4}$, $0 \leq r \leq \sqrt{2} \cos \theta$, 且此时 $\sqrt{2} \cos \theta \geq 1$.

$$\int_0^1 (1-r)r dr + \int_1^a (r-1)r dr = \frac{a^3}{3} - \frac{a^2}{2} + \frac{1}{3}, \quad a = \sqrt{2} \cos \theta.$$

于是

$$I = \int_0^{\frac{\pi}{4}} \left(\frac{2\sqrt{2}}{3} \cos^3 \theta - \cos^2 \theta + \frac{1}{3} \right) d\theta = \frac{2\sqrt{2}}{3} \cdot \frac{5\sqrt{2}}{12} - \left(\frac{\pi}{8} + \frac{1}{4} \right) + \frac{\pi}{12} = \frac{11}{36} - \frac{\pi}{24}.$$

28. 若 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 则由 $a(\alpha_1 + k\alpha_3) + b(\alpha_2 + \mu\alpha_3) = 0$ 得 $a = b = 0$, 充分性成立.

反之取 $\alpha_3 = 0$, α_1, α_2 线性无关, 则对任意 k, μ 结论成立, 但 $\alpha_1, \alpha_2, \alpha_3$ 线性相关, 必要性不成立. 选 B.

29. 消去 z : 第一式 $+3 \times$ 第二式得 $7x - y = 11$, 此即 L 关于 z 轴方向的投影柱面. 故

$$\text{在 } z = 0 \text{ 上的投影: } \begin{cases} 7x - y = 11, \\ z = 0; \end{cases} \quad \text{在 } z = 1 \text{ 上的投影: } \begin{cases} 7x - y = 11, \\ z = 1. \end{cases}$$

30. $\lambda y_1 + \mu y_2$ 为非齐次解 $\iff \lambda + \mu = 1$; $\lambda y_1 - \mu y_2$ 为齐次解 $\iff \lambda - \mu = 0$. 故 $\lambda = \mu = \frac{1}{2}$, 选 B.

31. 两边取行列式: $|A^*| = |A|^2$, $|-2A| = -8|A|$, 故 $|A|^2 = -8|A|$. 若 $|A| = 0$, 则 $r(A^*) \leq 1$, 与 $A^* = -2A \neq O$ 的秩相矛盾 (逐一验证 $r(A) = 1, 2$ 均不可能), 故 $|A| = -8$.

由 $A^* = |A|A^{-1} = -8A^{-1} = -2A$ 得 $A^2 = 4E$, 于是

$$\text{tr}[(A^2)^{-1}] = \text{tr}\left(\frac{1}{4}E\right) = \frac{3}{4}.$$

32. 令 $u = x^2 \geq 0$, $v = y^2 \geq 0$, $s = u + v$, 则 $|u - v| \leq s$,

$$|x^2 - y^2|e^{-x^2 - y^2} = |u - v|e^{-s} \leq se^{-s} \leq \frac{1}{e},$$

等号在 $s = 1$ 、 $uv = 0$ (如 $x = 1, y = 0$) 处取到. 故 $k_{\min} = \frac{1}{e}$.

33. $\frac{\partial P}{\partial y} = 6xy^2 - 2y \cos x = \frac{\partial Q}{\partial x}$, 场为保守场. 取势函数

$$u(x, y) = x^2 y^3 - y^2 \sin x + y,$$

$$\text{故 } W = u\left(\frac{\pi}{2}, 1\right) - u(0, 0) = \frac{\pi^2}{4} - 1 + 1 = \frac{\pi^2}{4}.$$

34. $\text{Cov}\left(\frac{3X+Y}{2}, \frac{X-2Y}{3}\right) = \frac{1}{6}(3\sigma^2 - 5\rho\sigma^2 - 2\sigma^2) = \frac{\sigma^2}{6}(1 - 5\rho) = 0 \implies \rho = \frac{1}{5}.$

35. $f(x) = -\ln(x + \sqrt{1+x^2})$, $f'(x) = -\frac{1}{\sqrt{1+x^2}} = -1 + \frac{x^2}{2} - \frac{3x^4}{8} + \dots$, 故

$$f(x) = -x + \frac{x^3}{6} - \frac{3x^5}{40} + \dots, \quad f^{(5)}(0) = 5! \cdot \left(-\frac{3}{40}\right) = -9.$$

36. (I) $f = (x_1 + x_3)^2 + 2x_2^2 + (a-1)x_3^2$, 化为 $y_1^2 + y_2^2$ 要求秩 2、正惯性指数 2, 故 $a = 1$.

取 $y_1 = x_1 + x_3$, $y_2 = \sqrt{2}x_2$, $y_3 = x_3$, 即

$$P = \begin{pmatrix} 1 & 0 & -1 \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad X = PY.$$

$$(II) A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix} \text{ 的特征值为 } 0, 2, 2, \text{ 故 } X^T X = 1 \text{ 时 } f_{\max} = 2;$$

$\lambda = 2$ 的特征向量为 $c_1(1, 0, 1)^T + c_2(0, 1, 0)^T$, 由 $x_1 = x_2 > 0$ 及单位化得最大值点 $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)^T$.

37. 参数化 $(1+t, -1+2t, 2+t)$, $t \in [0, 1]$, $ds = \sqrt{6}dt$, $x^2 + y^2 + z^2 = 6t^2 + 2t + 6$,

$$I = \sqrt{6} \int_0^1 (6t^2 + 2t + 6)dt = 9\sqrt{6}.$$

38. $V = 2 \iint_{r \leq R \cos \theta} \sqrt{R^2 - r^2} d\sigma = \frac{2}{3}R^3 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 - |\sin \theta|^3) d\theta = \frac{2}{3}R^3 \left(\pi - \frac{4}{3}\right) = \frac{2}{9}R^3(3\pi - 4).$

39. 由中心极限定理, $\hat{p}$ 近似服从 $N(0.8, \frac{0.16}{n})$,

$$P\{|\hat{p} - 0.8| < 0.04\} \approx 2\Phi\left(\frac{0.04\sqrt{n}}{0.4}\right) - 1 \geq 0.9 \implies 0.1\sqrt{n} \geq 1.645,$$

$n \geq 270.6$, 故至少生产 271 件.

40. (I) 分子分母同除 x^2 : $\frac{1 - \frac{\sin x}{x}}{1 + \frac{\sin \frac{1}{x}}{x}} \rightarrow 1$.

(II) $\sqrt[3]{abc}$.

(III) 分子 $= \ln(1 + e^{-x} \sin^2 x) \sim x^2$, 分母 $= \ln(1 - x^2 e^{-2x}) \sim -x^2$, 极限 $= -1$.

(IV) $(1+x)^{3/x} = e^3 e^{-\frac{3x}{2} + o(x)}$, 极限 $= -\frac{3e^3}{2}$.

(V) $\frac{e^x(e^{\tan x - x} - 1)}{x^3} \sim \frac{\tan x - x}{x^3} \rightarrow \frac{1}{3}$.

(VI) $\cot x \cdot \frac{x - \sin x}{x \sin x} \sim \frac{1}{x} \cdot \frac{x^3/6}{x^2} = \frac{1}{6}$.

(VII) 指数 $\frac{\ln(1 - x^2)}{1 - \sqrt{1 - x^2}} \rightarrow \frac{-x^2}{x^2/2} = -2$, 极限 $= e^{-2}$.

(VIII) $e^{\lim_{x \rightarrow 0} \sin x \ln x} = e^0 = 1$.

41. $BX = 0$ 的解总是 $ABX = 0$ 的解, 两者同解 $\iff$ 解空间维数相同 $\iff n - r(B) = n - r(AB)$, 即 $r(AB) = r(B)$. 选 B.

42. 当 $x > 0$ 时公比 $e^{-x} \in (0, 1)$ 收敛; $x = 0$ 时每项为 0, 和为 0; $x < 0$ 时发散. 收敛域 $[0, +\infty)$,

$$S(x) = \frac{x}{1 - e^{-x}} (x > 0), \quad S(0) = 0.$$

43. 由 $xf(x) = \frac{1}{\sqrt{1-x^2}}$ 得 $\frac{1}{f(x)} = x\sqrt{1-x^2}$, 故

$$\int \frac{dx}{f(x)} = \int x\sqrt{1-x^2} dx = -\frac{1}{3}(1-x^2)^{\frac{3}{2}} + C.$$

选 C.

44. $P\{Z = 1\} = P\{X \leq Y\} = \int_0^{+\infty} \lambda_1 e^{-\lambda_1 x} e^{-\lambda_2 x} dx = \frac{\lambda_1}{\lambda_1 + \lambda_2}, \quad P\{Z = 0\} = \frac{\lambda_2}{\lambda_1 + \lambda_2}.$

$$F_Z(z) = \begin{cases} 0, & z < 0, \\ \frac{\lambda_2}{\lambda_1 + \lambda_2}, & 0 \leq z < 1, \\ 1, & z \geq 1. \end{cases}$$

45. 由 $\limsup \sqrt[n]{|a_n|} = 1$ 及 $\sqrt[n]{n!} \rightarrow +\infty$ 得 $\limsup \sqrt[n]{|a_n|/n!} = 0$, 故收敛半径为 $+\infty$, 收敛域为 $(-\infty, +\infty)$.

46. 曲面 $x^2 + y^2 = 5$ 在 P 处外法向 $\mathbf{n} = (2x, 2y, 0)|_P = (2, 4, 0)$, 单位化 $\frac{1}{\sqrt{5}}(1, 2, 0)$;

$$\nabla u = (y^2 z^3, 2xyz^3, 3xy^2 z^2)|_P = (-4, -4, 12),$$

$$\frac{\partial u}{\partial \mathbf{n}} = \frac{-4 - 8}{\sqrt{5}} = -\frac{12\sqrt{5}}{5}.$$

47. 底面 $z = 0$ 贡献 0 ; 顶面 $z = x + R$, $dS = \sqrt{2} dx dy$:

$$\iint_{x^2+y^2 \leq R^2} (x+R)\sqrt{2} dx dy = \sqrt{2}\pi R^3;$$

侧面 $x = R \cos \theta, y = R \sin \theta, 0 \leq z \leq R(1 + \cos \theta)$, $dS = R d\theta dz$:

$$\int_0^{2\pi} \int_0^{R(1+\cos \theta)} z R dz d\theta = \frac{R^3}{2} \int_0^{2\pi} (1 + \cos \theta)^2 d\theta = \frac{3\pi}{2} R^3.$$

$$\text{故 } I = \left(\sqrt{2} + \frac{3}{2} \right) \pi R^3.$$

48. 记 $C = A - B \neq O$, $C^2 = O$ 知特征值全为 0, 且 $r(C) + r(C) \leq 3$, 故 $r(C) = 1$, 线性无关特征向量个数 $= 3 - r(C - 0E) = 2$. 选 C.

49. 特征方程 $\lambda^2 + a\lambda + 1 = 0$, 两根之积为 1.

$a > 2$: 两负实根; $a = 2$: 二重根 -1 ; $0 < a < 2$: 共轭复根实部 $-\frac{a}{2} < 0$; $a = 0$: $\pm i$, 解为 $\cos x, \sin x$ 有界.

$a < 0$ 时实部 (或实根) 为正, 解无界. 故 $a \in [0, +\infty)$.

50. $H_0: \mu \geq 12$, $H_1: \mu < 12$, σ 已知用 U 检验:

$$u = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{11.2 - 12}{2.6/10} \approx -3.08 < -1.645,$$

落入拒绝域, 故在 $\alpha = 0.05$ 下不能认为该批圆木小头平均直径在 12 cm 以上.

51. (I) 由对称性 $\bar{x} = \bar{y} = 0$; V 的体积 $= \int_0^1 \pi z dz = \frac{\pi}{2}$, $\iiint z dV = \int_0^1 \pi z^2 dz = \frac{\pi}{3}$, 故质心为 $(0, 0, \frac{2}{3})$.

(II) Σ 在 $M(a, b, a^2 + b^2)$ 处切平面为 $2ax + 2by - z - (a^2 + b^2) = 0$, 记 $t = a^2 + b^2 \in [0, 1]$,

$$d(t) = \frac{t + \frac{2}{3}}{\sqrt{4t+1}}, \quad d'(t) = \frac{2t - \frac{1}{3}}{(4t+1)^{3/2}},$$

故 $t = \frac{1}{6}$ 时最近, $L: x^2 + y^2 = \frac{1}{6}, z = \frac{1}{6}$. 在 L 上 $dz = 0$,

$$I = \oint_L x dy - y dx = 2 \cdot \text{面积} = 2\pi \cdot \frac{1}{6} = \frac{\pi}{3}.$$

52. 由 $m \frac{dv}{dt} = -kv$ 及 $\frac{dv}{dt} = v \frac{dv}{ds}$ 得 $m \frac{dv}{ds} = -k$, 即 $v = v_0 - \frac{k}{m}s$.

取 $v_0 = 600 \text{ km/h} = \frac{500}{3} \text{ m/s}$, $v_1 = 100 \text{ km/h} = \frac{250}{9} \text{ m/s}$, $s = 500 \text{ m}$:

$$k = \frac{m(v_0 - v_1)}{s} = \frac{4500 \times \frac{1250}{9}}{500} = 1250 \text{ (kg/s)}.$$

飞机滑行至停止的总距离 $s_{\max} = \frac{mv_0}{k} = \frac{4500 \times \frac{500}{3}}{1250} = 600 \text{ m}$, 故跑道至少 600 m.

53. 设 A 的特征值为 λ , 则 A^* 的特征值为 $\frac{|A|}{\lambda} = \frac{2}{\lambda}$, 由 $A^* = A - E$ 得 $\frac{2}{\lambda} = \lambda - 1$, 即 $\lambda^2 - \lambda - 2 = 0$, $\lambda = 2$ 或 -1 .

又 $\lambda_1 \lambda_2 \lambda_3 = |A| = 2$, 只能是 $2, -1, -1$, 正、负惯性指数分别为 1, 2. 选 D.

54. 用拉格朗日乘数法: $2 = \lambda(8x - 2y)$, $1 = \lambda(8y - 2x) \Rightarrow y = \frac{2}{3}x$,

代入约束得 $\frac{40}{9}x^2 = \frac{5}{8}$, $x = \pm \frac{3}{8}$, $2x + y = \frac{8}{3}x = \pm 1$.

故 $2x + y \in [-1, 1]$.

55. (I) 记行、列交换的初等矩阵为 P, Q , 则 $B = PAQ$, $A = PBQ$:

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}, \quad \alpha_1 = (1, 0, 1)^T, \alpha_2 = (1, 1, -1)^T, \alpha_3 = (1, -1, 1)^T.$$

(II) 施密特正交化: $k = \frac{(\alpha_2, \beta_1)}{(\beta_1, \beta_1)} = 0$, $l_1 = \frac{(\alpha_3, \beta_1)}{(\beta_1, \beta_1)} = 1$, $l_2 = \frac{(\alpha_3, \beta_2)}{(\beta_2, \beta_2)} = -\frac{1}{3}$,
此时 $\beta_2 = (1, 1, -1)^T$, $\beta_3 = \frac{1}{3}(1, -2, -1)^T$.

(III) 单位化得

$$Q = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{pmatrix}, \quad T = Q^T A = \begin{pmatrix} \sqrt{2} & 0 & \sqrt{2} \\ 0 & \sqrt{3} & -\frac{1}{\sqrt{3}} \\ 0 & 0 & \frac{\sqrt{6}}{3} \end{pmatrix}.$$

56. 记 $f(x) = \sum a_n x^n$, 则条件即 $2xf(x) + f'(x) = 0$, 解得 $f(x) = Ce^{-x^2}$; 由 $a_0 = f(0) = 1$ 得 $f(x) = e^{-x^2}$.

故 $\sum_{n=0}^{\infty} a_n = f(1) = e^{-1}$.

57. 条件即 $2 \leq \frac{x}{x^2 + y^2} \leq 4$, $2 \leq \frac{y}{x^2 + y^2} \leq 4$. 作反演变换

$$u = \frac{x}{x^2 + y^2}, \quad v = \frac{y}{x^2 + y^2}, \quad |J| = \frac{1}{(x^2 + y^2)^2}, \quad uv = \frac{xy}{(x^2 + y^2)^2},$$

则 $\frac{1}{xy} dx dy = \frac{1}{uv} du dv$, 于是

$$I = \int_2^4 \int_2^4 \frac{du dv}{uv} = (\ln 2)^2.$$

58. (I) 由前两式 $x_1 = x_4 - 2$, $x_2 = x_4 - 4$, 代入第三式得 $x_3 = 2x_4 - 5$, 故

$$X = (-2, -4, -5, 0)^T + t(1, 1, 2, 1)^T.$$

(II) 将通解代入 ② 各式并令对 t 恒成立:

$$(a-2)t + (3-4a) = -5 \Rightarrow a = 2; (b-4)t + (5-4b) = -11 \Rightarrow b = 4; -5 = -c + 1 \Rightarrow c = 6.$$

此时 ② 的系数矩阵秩为 3, 解空间维数也是 1, 两方程组同解. 故 $a = 2, b = 4, c = 6$.

59. A: $\frac{1}{x^2\sqrt{1+x}} \sim x^{-5/2}$, $\frac{5}{2} > 1$ 收敛.

B: $\frac{1}{\ln(1+x)} \sim \frac{1}{x}$ 在 0 附近发散; C: $\frac{1}{\sin x} \sim \frac{1}{x}$ 发散; D: 被积函数趋于 ± 1 , 发散. 选 A.

60. (I) 令 $F(x) = x^2 f(x)$, $F(a) = F(b) = 0$, 由罗尔定理存在 $\xi \in (a, b)$ 使

$$F'(\xi) = \xi[2f(\xi) + \xi f'(\xi)] = 0,$$

因 $\xi > a > 0$, 故 $2f(\xi) + \xi f'(\xi) = 0$.

(II) 令 $G(x) = e^{-x^2} f(x)$, $G(a) = G(b) = 0$, 由罗尔定理存在 $\eta \in (a, b)$ 使

$$G'(\eta) = e^{-\eta^2} [f'(\eta) - 2\eta f(\eta)] = 0 \Rightarrow 2\eta f(\eta) - f'(\eta) = 0.$$

61. (I) z 轴与 L 交于原点, 夹角余弦 $\frac{1}{\sqrt{3}}$, 旋转面为圆锥: 点 (x, y, z) 满足

$$\frac{(x+y+z)^2}{3} = \frac{x^2+y^2+z^2}{3} \cdot 3 \cdot \frac{1}{3} \implies (x+y+z)^2 = x^2+y^2+z^2,$$

即 $\Sigma: xy + yz + zx = 0$.

(II) 在平面 $x+y+z=1$ 上, L_1 满足 $x^2+y^2+z^2=1$, 是圆心 $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ 、半径 $\sqrt{\frac{2}{3}}$ 的圆.

$\text{rot}(z, x, y) = (1, 1, 1)$, 由斯托克斯公式

$$I = \iint_S (1, 1, 1) \cdot \frac{(1, 1, 1)}{\sqrt{3}} dS = \sqrt{3} \cdot \pi \cdot \frac{2}{3} = \frac{2\sqrt{3}}{3} \pi.$$

62. $E+B = (E+A)^{-1}[(E+A) + (E-A)] = 2(E+A)^{-1} \implies [(E+B)^2]^{-1} = \frac{1}{4}(E+A)^2.$

$$E+A = \begin{pmatrix} 2 & 0 & 0 \\ 2 & 4 & 0 \\ 0 & 4 & 6 \end{pmatrix}, \quad (E+A)^2 = \begin{pmatrix} 4 & 0 & 0 \\ 12 & 16 & 0 \\ 8 & 40 & 36 \end{pmatrix}, \quad [(E+B)^2]^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 4 & 0 \\ 2 & 10 & 9 \end{pmatrix}.$$

63. (I) 令 $g(x) = f(x + \frac{1}{2}) - f(x)$, $x \in [0, \frac{1}{2}]$, 则

$$g(0) = f(\frac{1}{2}) - f(0), \quad g(\frac{1}{2}) = f(1) - f(\frac{1}{2}) = -g(0).$$

若 $g(0) = 0$, 取 $\xi = \frac{1}{2} \in (0, 1)$ 即可; 否则 $g(0), g(\frac{1}{2})$ 异号, 由介值定理存在 $\xi \in (0, \frac{1}{2})$ 使 $g(\xi) = 0$.

(II) 令 $g(x) = f(x + \frac{1}{n}) - f(x)$, $x \in [0, 1 - \frac{1}{n}]$, 则

$$\sum_{k=0}^{n-1} g\left(\frac{k}{n}\right) = f(1) - f(0) = 0.$$

若某个 $g(\frac{k}{n}) = 0$ ($k \geq 1$) 即得; 否则其中必有异号者, 由介值定理得 $\xi \in (0, 1 - \frac{1}{n}) \subset (0, 1)$ 使 $f(\xi) = f(\xi + \frac{1}{n})$.

64. A 不单调不减; C 中 $F(+\infty) = \frac{1}{2} \neq 1$; D 未设 $f \geq 0$, 不能保证单调不减.

B 中 F 连续、单调增, $F(-\infty) = 0, F(+\infty) = 1$. 选 B.

65. $z = \frac{2x}{x^2 - y^2} = \frac{1}{x-y} + \frac{1}{x+y}, \quad \frac{\partial^n z}{\partial y^n} = \frac{n!}{(x-y)^{n+1}} + \frac{(-1)^n n!}{(x+y)^{n+1}}.$

在 $(2, 1)$ 处 $x-y=1, x+y=3$, 故

$$\frac{\partial^n z}{\partial y^n} \Big|_{(2,1)} = n! \left[1 + \frac{(-1)^n}{3^{n+1}} \right].$$

66. $u_n \rightarrow 0$, 当 n 充分大时 $0 < u_n < 1$, 故 $u_n^2 < u_n$, 由比较判别法 $\sum u_n^2$ 收敛.

B 取 $u_n = \frac{1}{n^2}$ 反例; C、D 中极限未必存在 (可等于 1). 选 A.

67. 齐次解 $C_1 \cos x + C_2 \sin x$; $y'' + y = x$ 的特解 x ; $y'' + y = \cos x$ 发生共振, 特解 $\frac{x}{2} \sin x$. 故

$$y = C_1 \cos x + C_2 \sin x + x + \frac{x}{2} \sin x.$$

68. 由 $AB \subset A, \overline{A} \overline{B} \subset \overline{A}$ 及二者相等知 $AB = \overline{A} \overline{B} = \emptyset$,

故 $A \cup B = \Omega$ 且 $AB = \emptyset$, 即 $B = \overline{A}$. 于是

$$P(\overline{A} | B) = \frac{P(B)}{P(B)} = 1, \quad P(A | \overline{B}) = \frac{P(A)}{P(A)} = 1, \quad \text{原式} = 2.$$

$$69. EU = EV = \frac{2}{\pi}, E(UV) = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} \sin x \cos x dx = \frac{1}{\pi},$$

$$\text{Cov}(U, V) = \frac{1}{\pi} - \frac{4}{\pi^2} = \frac{\pi - 4}{\pi^2} < 0,$$

且 U, V 不是线性关系, $|\rho| \neq 1$. 选 D.

$$70. x = \pm 1 \text{ 为两条铅直渐近线; } x \rightarrow -\infty \text{ 时 } y \rightarrow 0, \text{ 得水平渐近线 } y = 0;$$

$x \rightarrow +\infty$ 时 $\ln(1 + e^x) = x + \ln(1 + e^{-x}) \rightarrow x$, 得斜渐近线 $y = x$. 共 4 条, 选 D.

$$71. P = \frac{1}{y} + yf(xy), Q = xf(xy) - \frac{x}{y^2}, \text{ 验证 } \frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x} = f(xy) + xyf'(xy) - \frac{1}{y^2}, \text{ 积分与路径无关.}$$

取 $F' = f$, 势函数 $u = \frac{x}{y} + F(xy)$. 由于 A, B 两点均有 $xy = 2$,

$$I = u(1, 2) - u\left(3, \frac{2}{3}\right) = \frac{1}{2} - \frac{9}{2} = -4.$$

$$72. E\bar{X}^2 = D\bar{X} = \frac{1}{n}, ES^2 = 1, \text{ 又 } \bar{X} \text{ 与 } S^2 \text{ 独立故 } E(\bar{X}S) = E\bar{X} \cdot ES = 0,$$

$$E(T^2) = n[E\bar{X}^2 - 2E(\bar{X}S) + ES^2] = n\left(\frac{1}{n} + 1\right) = n + 1.$$

$$73. (\text{I}) \text{ 由 } A\alpha = \lambda\alpha \text{ 三个分量: } 2 - 1 - 2 = \lambda, 5 + a - 3 = \lambda, -1 + b + 2 = -\lambda,$$

得 $\lambda = -1, a = -3, b = 0$.

(II) 此时 $\text{tr}A = -3, |A| = -1$, 结合 $\lambda = -1$ 知特征值为三重 -1 . 而

$$A + E = \begin{pmatrix} 3 & -1 & 2 \\ 5 & -2 & 3 \\ -1 & 0 & -1 \end{pmatrix}, \quad r(A + E) = 2,$$

特征向量只有 $3 - 2 = 1$ 个线性无关, 故 A 不能相似于对角矩阵.

$$74. X, Y \text{ 独立且同服从 } N(0, \frac{1}{2}), \text{ 故 } X - Y \sim N(0, 1), E|X - Y| = \sqrt{\frac{2}{\pi}},$$

$$P\{X - Y > \sqrt{2/\pi}\} = 1 - \Phi\left(\sqrt{\frac{2}{\pi}}\right).$$

选 C.

$$75. f'(t) = \int_0^t e^{-(x-t)^2} dx, \quad f''(t) = 1 + \int_0^t 2(x-t)e^{-(x-t)^2} dx = 1 + (e^{-t^2} - 1) = e^{-t^2},$$

故 $f''(1) = e^{-1}$.